

Kuangshi (Leo) Ai

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Education

University of Notre Dame

Ph.D. Student in Computer Science and Engineering

Indiana, USA

2024.08-Present

- **Research Interests:** Scientific Visualization, Multi-agent LLM Systems, Human-Computer Interaction
- **GPA:** 4.0/4.0

Fudan University

B.E. in Artificial Intelligence, School of Data Science

Shanghai, China

2020.08-2024.07

- **GPA:** 3.71/4.0 **Rank:** 2/16

The University of Sydney

Exchange Student in the Faculty of Engineering

Sydney, Australia

2023.02-2023.06

- **Academic Performance:** High Distinction **U.S. GPA Equivalent:** 4.0/4.0

Publications

- **Kuangshi Ai**, Kaiyuan Tang, Chaoli Wang. NLI4VolVis: Natural Language Interaction for Volume Visualization via Multi-LLM Agents and Editable 3D Gaussian Splatting. *In Proceedings of IEEE Transactions on Visualization and Computer Graphics (IEEE VIS), 2025. Best Paper Award (Top 1%)*
- Kaiyuan Tang, **Kuangshi Ai**, Jun Han, Chaoli Wang. TexGS-VolVis: Expressive Scene Editing for Volume Visualization via Textured Gaussian Splatting. *In Proceedings of IEEE Transactions on Visualization and Computer Graphics (IEEE VIS), 2025*
- **Kuangshi Ai**, Haichao Miao, Zhimin Li, Chaoli Wang, Shusen Liu. An Evaluation-Centric Paradigm for Scientific Visualization Agents. *1st Workshop on GenAI, Agents, and the Future of VIS (IEEE VIS), 2025*
- Simret Araya Gebreegziabher, **Kuangshi Ai**, Zheng Zhang, Elena Glassman, Toby Jia-Jun Li. Leveraging Variation Theory in Counterfactual Data Augmentation for Optimized Active Learning. *In Proceedings of Findings of the Association for Computational Linguistics (ACL), 2025*
- **Kuangshi Ai**^{*}, Jonathan A. Karr Jr^{*}, Meng Jiang, Nitesh V. Chawla, Chaoli Wang. KEO: Knowledge Extraction on OMIn via Knowledge Graphs and RAG for Safety-Critical Aviation Maintenance. *Preprint, 2025* (* indicates co-first authors)

Research Experience

Natural Language Interaction for Volume Visualization

Instructor: Prof. Chaoli Wang

University of Notre Dame, USA

2024.08-Present

- Project Website: <https://nli4volvis.github.io>
- Developed the first-ever framework enabling natural language interaction with complex 3D scenes using LLM agents and editable 3D Gaussian splatting
- Integrated vision-language foundation models to achieve open-vocabulary querying of objects in 3D volumetric scenes, supporting intuitive scene manipulation with natural language
- **Best Paper Award** at IEEE VIS 2025

Counterfactual Data Augmentation for Active Learning

Instructor: Prof. Toby Li

University of Notre Dame, USA

2023.07-2024.05

- Code available at: <https://github.com/KuangshiAi/VariationTheory>
- Inspired by Variation Theory, proposed an Active Learning (AL) approach which synthesizes artificial datapoints that highlight key similarities and differences among labels
- Developed a neuro-symbolic pipeline combining large language models and rule-based models, which generates high-quality counterfactual examples
- Achieved a comparable accuracy to prevalent AL strategies while necessitating fewer annotations

Data-Efficient Preprocessing of Vision Transformers

The University of Sydney, Australia

Instructor: Prof. Chang Xu

2023.03-2023.06

- Code available at: https://github.com/KuangshiAi/vision_transformer
- Proposed and implemented a method to preprocess and regenerate visual data input sequences, achieving minimal throughput loss and performance improvements on four small-scaled datasets
- Pointed out potentials of ViTs and the limitation of our method on large-scaled datasets, and proposed a practical guideline for ViT training

Bilevel Optimization with Special Forms of Lower-level Functions

Fudan University, China

Instructor: Prof. Rujun Jiang

2022.12-2023.07

- Conducted experiments on IJCNN1 and corrupted MNIST datasets to evaluate stochastic and variance reduction algorithms for strongly convex lower-level functions
- Proposed case-specific algorithms for typical machine learning problems based on the structure of lower-level functions

*Internship Experience***Quantitative Research Intern**

Beijing, China

Ningbo Lingjun Investment Management Partnership

2024.01-2024.05

- Participated in the development and optimization process of the combo model, maintaining and proficiently mastering the company's live trading model framework
- Based on the company's factor library and financial data, designed a graph transformer-based daily-frequency investment portfolio model, which demonstrated superior effectiveness and robustness compared to the existing live trading model through backtesting.

Embodied AI Research Intern

Shanghai, China

Shanghai Ziyang Intelligent Technology Co., Ltd.

2023.10-2023.12

- As an early member of the startup, developing autonomous control systems for humanoid robots to enhance their capabilities for everyday tasks
- Combined the structure of Swin ViT and Resformer to achieve robot vision focusing ability, and developed a Fast-ACVNet based stereo depth estimation pipeline for affordable environmental information capture
- Deployed and fine-tuned RT2 and Qwen locally as the high-level control unit of robot dogs, and leveraged GPT-4-vision to build navigation system in Meta's Habitat simulation environment

Business Operations and Data Management Intern

Yunnan, China

YCIH Logistic Co., Ltd

2022.07-2022.08

- Utilized PostgreSQL database management system and Python web crawler technology to collect, analyze and manage data for the company's "INJA" Intelligent Supply Chain Platform
- Developed a backend system for data monitoring and control, real-time data visualization, and data analysis modeling

Honors & Awards

- **Best Paper Award** (Top 1%), IEEE VIS Conference, 2025
- **Outstanding Graduate**, Fudan University, 2024
- **Second Prize** (Top 2 in major), the Scholarship for Outstanding Students, Fudan University, 2024 & 2021
- **Third Prize**, the Scholarship for Outstanding Students, Fudan University, 2022
- **Outstanding Student Leaders**, Fudan University, 2020-2022
- **Second Prize in Shanghai**, Contemporary Undergraduate Mathematical Contest in Modeling, 2022
- **Scholarship for Outstanding Freshmen**, Fudan University, 2020

*Professional Skills***Programming:** Python, MATLAB, C/C++, SQL, R**Tools:** Pytorch, TensorFlow, paddlepaddle, LaTeX, Numpy, Pandas, Linux**Language:** English (TOEFL: 106/120), Chinese (native speaker)